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Define  $\bar{X}_n = \frac{1}{n} \sum_i w_i$

$$\left. \begin{aligned} E[\bar{X}_n] &= \mu_{\bar{X}_n} \\ \text{Var}(\bar{X}_n) &= \sigma_{\bar{X}_n}^2 \end{aligned} \right\} \frac{\bar{X}_n - \mu_{\bar{X}_n}}{\sqrt{\sigma_{\bar{X}_n}^2}} \xrightarrow{d} N(0,1) \text{ by CLT}$$


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$Z_n = \frac{1}{n} \sum_{i=1}^n X_i^2 \leftarrow \text{still a sample mean}$   $\{X_i\}_{i=1}^{\infty}$

$E(X_i) = \mu$   
 $\text{Var}(X_i) = \sigma_x^2$

$$E[Z_n] =$$

$$\text{Var}(Z_n) = \frac{Z_n - E(Z_n)}{\sqrt{\text{Var}(Z_n)}} \xrightarrow{d} N(0,1)$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\bar{x} \neq E[X_i] = \mu_x$$



WLLN

$$\bar{X}_n \xrightarrow{p} E[X_i]$$

The sample average

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \text{ is a random variable}$$

where  $\{X_i\}_{i=1}^n$  is a set of  
random variables

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) \xrightarrow{P} \text{Cov}(X_i, Y_i)$$

If  $(X_i, Y_i)_{i=1}^n$  i.i.d. vectors

$$= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})Y_i = \frac{1}{n} \sum X_i Y_i - \bar{X} \left( \frac{1}{n} \sum_{i=1}^n Y_i \right)$$

$$= \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y}$$

$$\frac{1}{n} \sum X_i Y_i \xrightarrow{P} E(X_i Y_i) \text{ by WLLN}$$

$$\xrightarrow{P} E(X_i Y_i) - E(X_i)E(Y_i) = \text{Cov}(X_i, Y_i)$$

$$\begin{aligned} \bar{X} &\xrightarrow{P} E(X_i) \\ \bar{Y} &\xrightarrow{P} E(Y_i) \end{aligned} \text{ by WLLN}$$

$$g: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$$

$$g(x, y) = xy$$

$$g(\bar{X}, \bar{Y}) = \bar{X} \bar{Y} \xrightarrow{P} g(E(X_i), E(Y_i)) = E(X_i)E(Y_i)$$

$$T_n = \frac{3.5 - 3}{\sqrt{10/1000}}$$

$\sim N(0,1)$   
 $\downarrow$  p-value

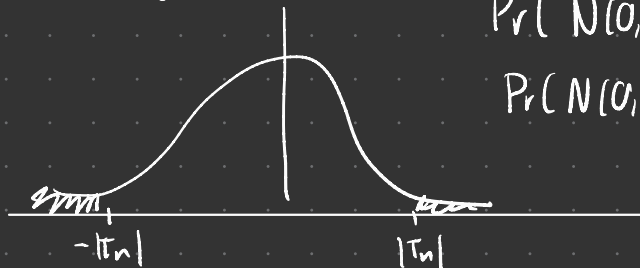
$$\Pr(|N(0,1)| > |T_n|) =$$

$$\bar{X}_n = 3.5$$

$$\mu_0 = 3$$

$$\hat{\sigma}_n^2 = 10$$

$$n = 1000$$



$$\Pr(N(0,1) > T_n) + \Pr(N(0,1) < -T_n)$$